Rabeya Jamshad

Computer Science and Engineering
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Research Interests

Robotics, Human-Robot Interaction, Artificial Intelligence, Reinforcement Learning, Healthcare Technology, Action Teams, Team Dynamics

Education

University of California San Diego 2021 - 2027 (Expected)
Ph.D., Computer Science and Engineering
Advisor: Dr. Laurel Riek

Georgia Institute of Technology 2017 - 2019
M.Sc., Electrical and Computer Engineering
Advisor: Dr. Ayanna Howard

Lahore University of Management Sciences 2011 - 2015
B.S., Electrical Engineering
Advisor: Dr. Abubakr Muhammad

Selected Publications

Jamshad, R., Pal, M., Gupta, V., and Riek, L. D. (2026). Title anonymized. *Planned Submission to RA-L.*

Jamshad, R., Ramaraj, P., Masanam, D., & Riek, L. D. (2026). Title anonymized. *ACM/IEEE International Conference on Human Robot Interaction (HRI 2027).* In *Review.*

Jamshad, R., Haripriyan, A., Sonti, A., Simkins, S. and Riek, L.D., (2026). Human-robot action teams: How robot proactivity affects teamwork. In *Computers in Human Behavior: Artificial Humans (CHBAH).*

Haripriyan, A., **Jamshad, R.,** Ramaraj, P., and Riek, L.D. (2024) Human-Robot Action Teams: A Behavioral Analysis of Team Dynamics. In *Proceedings of the 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN).*

Jamshad, R., Haripriyan, A., Sonti, A., Simkins, S., and Riek, L.D. (2024). Human-Robot Action Teams: How Robots Can Be Proactive Teammates. In *Companion Proc. of the ACM/IEEE International Conference on Human-Robot Interaction (HRI).*

Matsumoto, S., Ghosh, P., **Jamshad, R.,** and Riek, L. D. (2023). Robot, Uninterrupted: Telemedical Robots to Mitigate Care Disruption. In *Proceedings of the ACM/IEEE International Conference on Human Robot Interaction (HRI)*. [Acceptance rate: 25%]

Jamshad, R., Fry, K.E., Chen, Y.P. and Howard, A., (2019). Design of a robotic crib mobile to support studies in the early detection of cerebral palsy: A pilot study. In *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN).*

Experience

Ph.D. Student Researcher, UC San Diego 2021 - Present
Designed and developed proactive robot systems to support human-robot teaming

- Developed and validated new intent inference and behavior adaptation models for proactive robots to support fluent collaboration in human-robot teams
- Led and conducted multiple large-scale studies with humans and physical robots to validate new robot roles and behaviors in action teams
- Collected and analyzed multimodal human-robot interaction data

Senior Robotics Engineer, Xavor Corporation 2020 - 2021
Developed control system and SLAM for a mobile service robot
- Developed and integrated a multimodal AI control system in ROS for a companion robot for older adults

M.Sc. Student Researcher, Georgia Institute of Technology 2018 - 2019
Developed a robot for infants with neuromuscular disorders
- Developed a robotic crib mobile to facilitate the early diagnosis of neuromuscular disorders in infants
- Validated the effectiveness of the robotic intervention through at-home experiments

Academic Service

Reviewer: THRI, InGroup, HRI, SGR, ICSR, UIST 2022 - 2026

Student Mentoring

Graduate and Undergraduate Mentor, UC San Diego 2023 - 2026
- Advika Sonti (B.S. Cognitive Science)
- Oviya Senthil (B.S. Computer Science)
- Priyanshu Arora (M.Sc. Computer Science)
- Arthi Haripriyan (M.Sc. Computer Science)
- Deep Masanam (Ph.D. Computer Science)

Teaching

Teaching Assistant - Human-Robot Interaction, UC San Diego Spring 2026
- Advised 20+ student project groups
- Led the hardware setup and verification for class robotics labs
- Graded labs, homeworks, and projects
- Led office hours to help clarify student concepts, support class progress, and identify appropriate methods for designing, building and validating robotic systems

Leadership

Voting Graduate Student Member on the Chancellor's Advisory Committee on the Status of Women 2022 - 2024

Secretary for UC San Diego Robotics Graduate Student Organization 2022 - 2024

Panelist on United States Educational Foundation in Pakistan (USEFP) STEM Careers for Women 2020

Invited Talks and Demos

Academy of Management 2025, InGroup 2024 Symposium, HRI 2024 Poster, InGroup 2023 Poster, UC San Diego Open House 2022-2023

Awards and Honors

Powell Fellowship 2021
Fulbright Scholar 2017

Professional Competencies

Software and Programming Languages
C++, Python, ROS/ROS2, Matlab, Linux, Arduino, Numpy, PyTorch

Control Design and Robotics
Control Systems, Mobile Robotics, Search and Optimization, Artificial Intelligence, Sensor Fusion, Human-Robot Interaction, Reinforcement Learning, Hardware Design, Embedded Systems

Research Methods
Designing, conducting, and analyzing controlled experiments with human subjects; Qualitative data collection and analysis; Rapid prototyping